Chapter 1.2 Distributions

Rick Durrett Probability: Theory and Examples

Hu Jingwu

Built from the local Chapter 1 LLMwiki flash cards.

Roadmap for 1.2

Learning goal

Move between laws, CDFs, densities/mass functions, atoms, and transformations.

1. Distribution functions and probability laws.
2. Atoms as jumps and countability of discontinuities.
3. Normal tail estimates.
4. Probability integral transform.
5. Density transformations: monotone maps, lognormal, and squares.

Distribution Function Properties

Theorem

If $F(x) = \mathbb{P}(X \leq x)$, then:

$$F \text{ is nondecreasing}, \qquad \lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to \infty} F(x) = 1,$$

and F is right-continuous.

Pitfall

A CDF need not be left-continuous. Its jumps are atoms.

LLMwiki: `Distribution function properties; ID: durrett_ch1_distribution_function`

How to Prove CDF Properties

Proof idea

Translate analytic properties of F into event inclusions and continuity of probability.

Proof.

If $x \leq y$, then $\{X \leq x\} \subseteq \{X \leq y\}$, so $F(x) \leq F(y)$. For right-continuity, take $x_n \downarrow x$. Then

$$\{X \leq x_n\} \downarrow \{X \leq x\}.$$

By continuity from above,

$$F(x_n) = \mathbb{P}(X \leq x_n) \downarrow \mathbb{P}(X \leq x) = F(x).$$

The limits at $\pm\infty$ follow from $\{X \leq n\} \uparrow \Omega$ and $\{X \leq -n\} \downarrow \varnothing$. $\square$

LLMwiki: `Distribution function properties; ID: durrett_ch1_distribution_function`

Every Valid CDF Determines a Law

Theorem

Any function $F : \mathbb{R} \to [0,1]$ that is nondecreasing, right-continuous, and satisfies

$$F(-\infty) = 0, \qquad F(\infty) = 1$$

is the CDF of a unique probability measure on $\mathcal{B}(\mathbb{R})$.

- Define $\mu((a,b]) = F(b) - F(a)$.
- Extend from half-open intervals to the Borel sigma-field.

LLMwiki: `Every valid CDF determines a law; ID: durrett_ch1_cdf_to_law`

Atoms Are Jumps

Fact

For a real random variable with CDF F,

$$\mathbb{P}(X = x) = F(x) - F(x-), \qquad F(x-) = \lim_{y \uparrow x} F(y).$$

Since $\{X < x\} = \bigcup_{n \geq 1} \{X \leq x - 1/n\}$, continuity from below gives

$$\mathbb{P}(X < x) = \lim_{n \to \infty} F(x - 1/n) = F(x-).$$

Therefore

$$\mathbb{P}(X = x) = \mathbb{P}(X \leq x) - \mathbb{P}(X < x) = F(x) - F(x-).$$

LLMwiki: `Atoms are jumps of the CDF; ID: durrett_ch1_atoms_jumps`

Standard Normal Tail Estimate

Inequality

If χ is standard normal and $x > 0$, then Durrett's normal-tail estimate gives

$$(x^{-1} - x^{-3})e^{-x^2/2} \le \int_x^\infty e^{-y^2/2}\,dy \le x^{-1}e^{-x^2/2}.$$

Hence

$$\mathbb{P}(\chi \ge x) = \frac{1}{\sqrt{2\pi}}\int_x^\infty e^{-y^2/2}\,dy.$$

LLMwiki: `Standard normal tail estimate; ID: durrett_ch1_normal_tail_bound`

Probability Integral Transform

Fact

If X has continuous CDF F, then

$$F(X) \sim \mathrm{Unif}(0,1).$$

- Proof route: compute $\mathbb{P}(F(X) \leq y)$ through a quantile.
- Flat intervals require care, but continuity makes their probability contribution zero.

LLMwiki: `Probability integral transform; ID: durrett_ch1_probability_integral_transform`

Monotone Density Change of Variables

Theorem

Suppose X has density f on $[\alpha, \beta]$ and g is strictly increasing and differentiable. If $Y = g(X)$, then

$$f_Y(y) = \frac{f(g^{-1}(y))}{g'(g^{-1}(y))}\mathbf{1}_{\{g(\alpha)<y<g(\beta)\}}.$$

For $g(\alpha) < y < g(\beta)$,

$$\mathbb{P}(Y \le y) = \mathbb{P}(X \le g^{-1}(y)) = \int_{\alpha}^{g^{-1}(y)} f(x)\,dx.$$

Differentiate with respect to y and use $(g^{-1})'(y) = 1/g'(g^{-1}(y))$.

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.1

Let X and Y be random variables on $(\Omega, \mathcal{F}, P)$ and let $A \in \mathcal{F}$. Define $Z = X$ on A and $Z = Y$ on A^c. Show that Z is a random variable.

LLMwiki: `Piecewise random variable measurability; ID: exercise_method_piecewise_random_variable_measurability`

Exercise 1.2.1: Proof Skeleton

It is enough to check $\{Z \leq x\} \in \mathcal{F}$ for each $x \in \mathbb{R}$. Write

$$\{Z \leq x\} = \big(A \cap \{X \leq x\}\big) \cup \big(A^c \cap \{Y \leq x\}\big).$$

Each set on the right is measurable, so Z is measurable.

LLMwiki: `Piecewise random variable measurability; ID: exercise_method_piecewise_random_variable_measurability`

Exercise 1.2.2

Let χ be standard normal. Use Durrett's normal-tail estimate to find upper and lower bounds for $\mathbb{P}(\chi \geq 4)$.

LLMwiki: `Standard normal tail estimate; ID: durrett_ch1_normal_tail_bound`

Exercise 1.2.2: Proof Skeleton

Durrett's estimate with $x = 4$ gives

$$\left(\frac{1}{4} - \frac{1}{64}\right) e^{-8} \le \int_4^\infty e^{-y^2/2}\,dy \le \frac{1}{4} e^{-8}.$$

Therefore

$$\frac{15}{64\sqrt{2\pi}} e^{-8} \le \mathbb{P}(\chi \ge 4) \le \frac{1}{4\sqrt{2\pi}} e^{-8}.$$

LLMwiki: `Standard normal tail estimate; ID: durrett_ch1_normal_tail_bound`

Exercise 1.2.3

Show that a distribution function has at most countably many discontinuities.

LLMwiki: `Countability of CDF jumps; ID: exercise_method_countable_cdf_discontinuities`

Exercise 1.2.3: Proof Skeleton

Let $\Delta F(x) = F(x) - F(x-)$. For $m, n \geq 1$, set

$$D_{m,n} = \{x \in [-n, n] : \Delta F(x) > 1/m\}.$$

Each $D_{m,n}$ is finite, since finitely many jumps in it have total size at most 1 and each jump is bigger than $1/m$. Every discontinuity has $\Delta F(x) > 0$, so it lies in some $D_{m,n}$. Hence the discontinuity set is a countable union of finite sets.

LLMwiki: `Countability of CDF jumps; ID: exercise_method_countable_cdf_discontinuities`

Exercise 1.2.4

Suppose $F(x) = \mathbb{P}(X \leq x)$ is continuous. Show that $Y = F(X)$ is uniform on $(0, 1)$, i.e.

$$\mathbb{P}(Y \leq y) = y, \qquad 0 \leq y \leq 1.$$

LLMwiki: `Probability integral transform; ID: durrett_ch1_probability_integral_transform`

Exercise 1.2.4: Proof Skeleton

For $0 \le y \le 1$, use the generalized quantile

$$x_y = \sup\{x : F(x) \le y\}.$$

By monotonicity and continuity, $\{F(X) \le y\}$ has the same probability as a suitable half-line endpoint with CDF value y. Flat intervals carry no probability mass because F is constant there. Thus

$$\mathbb{P}(F(X) \le y) = y.$$

LLMwiki: `Probability integral transform; ID: durrett_ch1_probability_integral_transform`

Exercise 1.2.5

Suppose X has continuous density f and is supported on $[\alpha, \beta]$. Let g be strictly increasing and differentiable on (α, β). Find the density of $g(X)$.

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.5: Proof Skeleton

Let $Y = g(X)$. For $g(\alpha) < y < g(\beta)$,

$$F_Y(y) = \mathbb{P}(g(X) \le y) = \mathbb{P}(X \le g^{-1}(y)) = \int_{\alpha}^{g^{-1}(y)} f(x)\, dx.$$

Differentiating,

$$f_Y(y) = \frac{f(g^{-1}(y))}{g'(g^{-1}(y))},$$

and $f_Y(y) = 0$ outside $(g(\alpha), g(\beta))$.

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.6

Suppose X is normally distributed. Use the previous exercise to find the density of $\exp(X)$, the lognormal distribution.

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.6: Proof Skeleton

If $X \sim N(\mu, \sigma^2)$ and $Y = e^X$, then $g^{-1}(y) = \log y$ and $g'(g^{-1}(y)) = y$. Hence, for $y > 0$,

$$f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log y - \mu)^2}{2\sigma^2}\right],$$

and $f_Y(y) = 0$ for $y \leq 0$.

LLMwiki: `Monotone density change of variables; ID: exercise_method_density_change_of_variables`

Exercise 1.2.7

(i) If X has density f, compute the distribution function and density of X^2.

(ii) Specialize to X standard normal to get the chi-square density with one degree of freedom.

LLMwiki: `Density of a square transform; ID: exercise_method_square_transform_density`

Exercise 1.2.7: Proof Skeleton

Let $Y = X^2$. For $y \geq 0$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} f(x)\,dx.$$

Thus, for $y > 0$,

$$f_Y(y) = \frac{f(\sqrt{y}) + f(-\sqrt{y})}{2\sqrt{y}}.$$

If X is standard normal, then

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2} \mathbf{1}_{(0,\infty)}(y).$$

LLMwiki: `Density of a square transform; ID: exercise_method_square_transform_density`

Checklist: How to Attack 1.2 Problems

1. Need to verify a CDF? Check monotonicity, right-continuity, and endpoint limits.
2. Need an atom? Compute $F(x) - F(x-)$.
3. Need countability of jumps? Group jumps by size and bounded interval.
4. Need a transformed density? Compute the transformed CDF first, then differentiate.
5. Need a normal tail? Keep the exponential term and constants separate.

LLMwiki: `Chapter 1.2 flash cards and exercise method cards`

Mini Practice

1. If F is continuous and strictly increasing, show $\mathbb{P}(F(X) \leq y) = y$ directly using F^{-1}.
2. If X has density f, find the density of $aX + b$ for $a < 0$.
3. Show that a CDF can have only finitely many jumps larger than 0.01.
4. If $X \sim N(0, 1)$, use the tail bound to estimate $\mathbb{P}(|X| \geq x)$.